

# Separable Equations



## Order and verifying solutions

- 1 State the order of the differential equation  $y'' + 3y' - 2y = \sin x$ .
  - 2 State the order of  $\left(\frac{dy}{dx}\right)^3 + xy = 0$ , and note that order depends only on the highest derivative, not on any power of it.
  - 3 Verify that  $y = e^{2x}$  is a solution of  $y' - 2y = 0$ .
  - 4 Verify that  $y = \frac{1}{x}$  is a solution of  $y' + y^2 = 0$  for  $x \neq 0$ .
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## Recognizing and solving separable equations

- 5 Solve  $\frac{dy}{dx} = xy$ .
  - 6 Solve  $\frac{dy}{dx} = \frac{y^2}{x}$  for  $x > 0$ .
  - 7 Solve  $\frac{dy}{dx} = \frac{x+1}{y}$ .
  - 8 Solve  $\frac{dy}{dx} = 6x^2y^2$ .
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## Equilibrium solutions and qualitative behavior

- 9 Find the equilibrium (constant) solutions of  $\frac{dy}{dx} = y(3 - y)$ .
  - 10 For  $\frac{dy}{dx} = y(3 - y)$ , determine whether  $y = 0$  and  $y = 3$  are stable or unstable by checking the sign of  $\frac{dy}{dx}$  just above and below each.
  - 11 Find the equilibrium solutions of  $\frac{dy}{dx} = (y - 2)(y + 1)$  and classify each as stable or unstable.
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## General versus particular solutions

- 12 Find the general solution of  $\frac{dy}{dx} = 2x$ , then find the particular solution satisfying  $y(1) = 3$ .
- 13 Explain, in terms of the arbitrary constant  $C$ , why a differential equation typically has infinitely many solutions but an initial value problem typically has exactly one.

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### Initial value problems

- 14 Solve  $\frac{dy}{dx} = 3x^2y$ ,  $y(0) = 2$ .
- 15 Solve  $\frac{dy}{dx} = -2xy$ ,  $y(0) = 5$ .
- 16 Solve  $\frac{dy}{dx} = 4x^3y$ ,  $y(0) = -3$ .
- 17 Solve  $y' = -\frac{1}{2}y$ ,  $y(0) = 8$ .

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### Application: radioactive decay and population growth

- 18 A radioactive substance decays according to  $\frac{dA}{dt} = -kA$ . If  $A(0) = 100$  g and  $A(5) = 50$  g, find  $k$ .
- 19 A population grows according to  $\frac{dP}{dt} = 0.03P$ , with  $P(0) = 200$ . Find  $P(10)$ .
- 20 A bank account earns continuously-compounded interest,  $\frac{dA}{dt} = 0.05A$ , with  $A(0) = 1000$ . Find  $A(20)$  and the doubling time.

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### Application: Newton's Law of Cooling

- 21 Newton's Law of Cooling gives  $\frac{dT}{dt} = -k(T - T_s)$ , where  $T_s$  is the surroundings' temperature. Separate variables and integrate to find the general solution.
- 22 A cup of coffee at  $90^\circ \text{C}$  is left in a  $20^\circ \text{C}$  room. If  $T(10) = 60^\circ \text{C}$ , find  $k$ , then find  $T(20)$ .

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### Further practice: separable equations

- 23 Solve  $\frac{dy}{dx} = \frac{1+y^2}{1+x^2}$ .
- 24 Solve  $y' = y \cos x$ ,  $y(0) = 1$ .
- 25 Solve  $\frac{dy}{dx} = e^{x-y}$ .
- 26 Solve  $x \frac{dy}{dx} = y \ln y$  for  $x, y > 0$ ,  $y(1) = e$ .

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### Mixed review

- 27 Determine whether  $\frac{dy}{dx} = x + y$  is separable. Justify your answer.
- 28 Solve  $\frac{dy}{dx} = \frac{\cos x}{y}$ ,  $y(0) = -2$ .
- 29 A tank of water drains according to Torricelli's Law,  $\frac{dh}{dt} = -kh^{1/2}$ , where  $h$  is the water height. Separate variables and integrate.
- 30 For the model  $\frac{dy}{dx} = \frac{2x}{y}$  with  $y(0) = 3$ , find the particular solution and state its domain.